

Tonogenesis in Northern Pomo: Lexical tone from pitch-accent, not segmental contrasts

1. What is tonogenesis?

Tonogenesis is the development of lexical tone in a previously non-tonal language.

2. What triggers tonogenesis?

Generally, tone arises through segmental contrast neutralization - "mechanical differences in pitch" are reinterpreted as contrastive tones (Hombert, Ohala & Ewan 1979).

- e.g. Vietnamese (Haudricourt 1954)

- Phase 1: syllable final *-ʔ* vs. *-h* contrast develops into ‘rising’ vs. ‘falling’ contour with CV sylls developing ‘mid-level’
- Phase 2: syllable initial ‘voiceless’ vs. ‘voiced’ contrast splits the existing tones into ‘high’ vs. ‘low’ register

Original →		Tonogenesis Phase 1 →		Tonogenesis Phase 2	
pa ba	No Tone	pa 1 ba 1	Mid-Level	pa 1 pa 1	Mid-Level Mid-Falling
paʔ baʔ	No Tone	pa 1 ba 1	Rising	pa 1 pa 1	Rising Low-Glottalized
pah bah	No Tone	pa 1 ba 1	Falling	pa 1 pa 1	Low-Falling Broken

3. Are there other paths to tonogenesis?

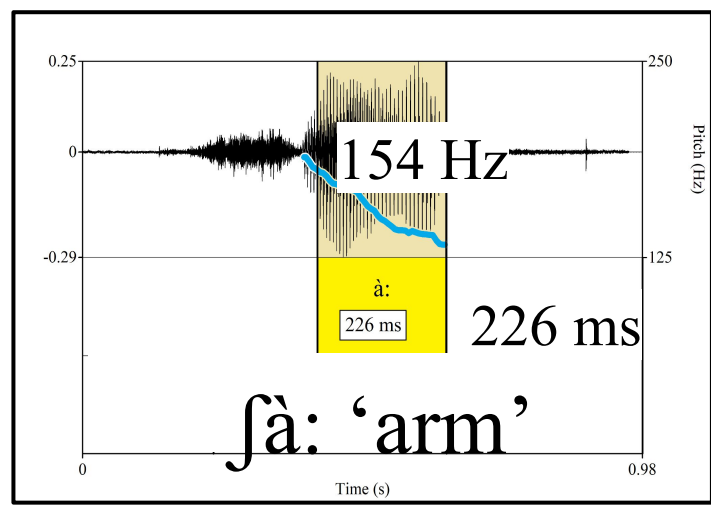
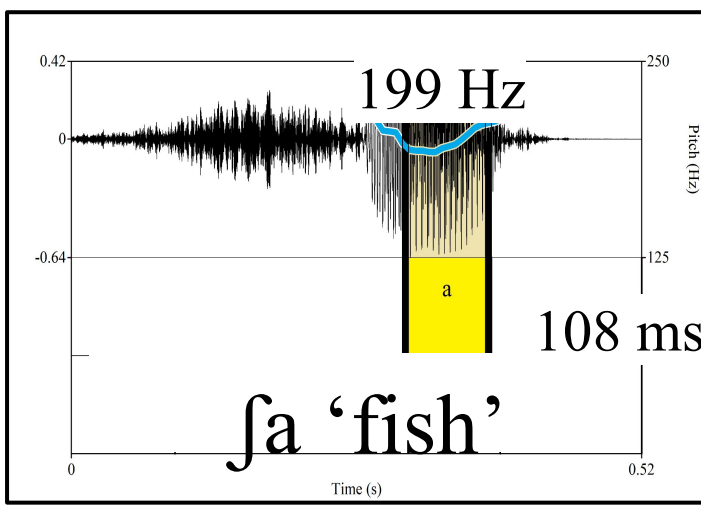
While segmental contrast neutralization is by far the most common path to tonogenesis, Northern Pomo recently developed tone with no evidence that it followed this path.

This paper argues that N. Pomo’s tones developed through the reanalysis of a pitch-accent.

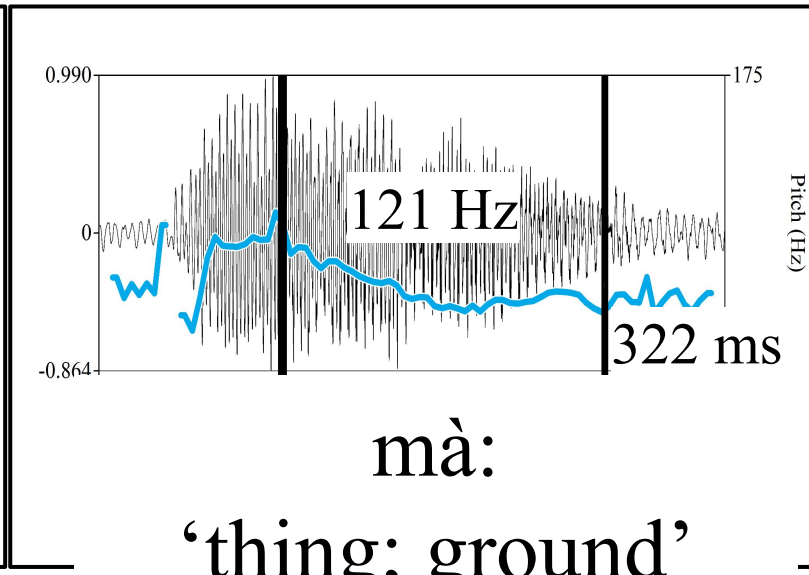
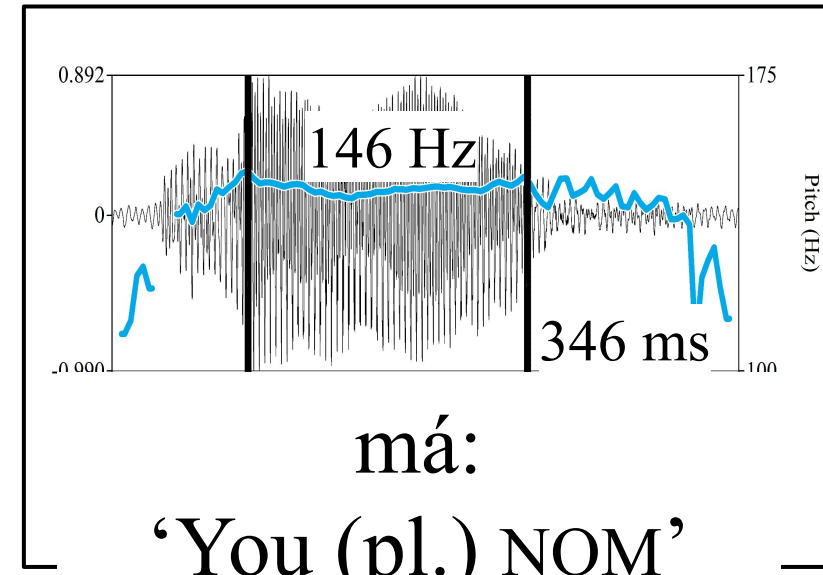
4. What does tone look like in Northern Pomo?

This description is based on the ~900 hrs of of field recordings made by Eero Vihman (1960s) and Cathy O’Connor (1979 - 2005) w/3 female speakers: Edna Campbell Guerrero, Annie Lake and Elenor Stevenson Gonzales.

- Words composed of CV syllables are associated with a **rising** pitch contour and words with CVV typically have a **L or falling** pitch contour:



- This makes pitch seem dependent on vowel length and led to the early description of “lexical pitch-accent” (McLendon 1973, Vihman 1976, O’Connor 1992).
- However, **minimal pairs** show a contrast between H vs. L on heavy syllables:



N. Pomo is not a canonical tone language - Tone is still heavily associated with vowel length. But...

**Northern Pomo has tonal minimal pairs on some heavy syllables!
We need an explanation of how those minimal pairs formed!**

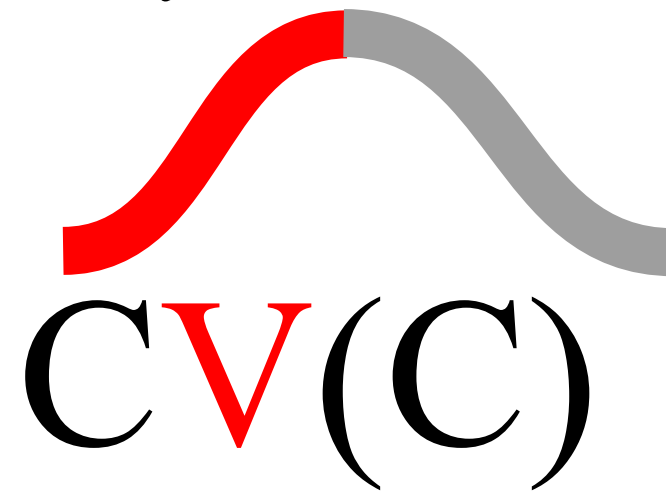
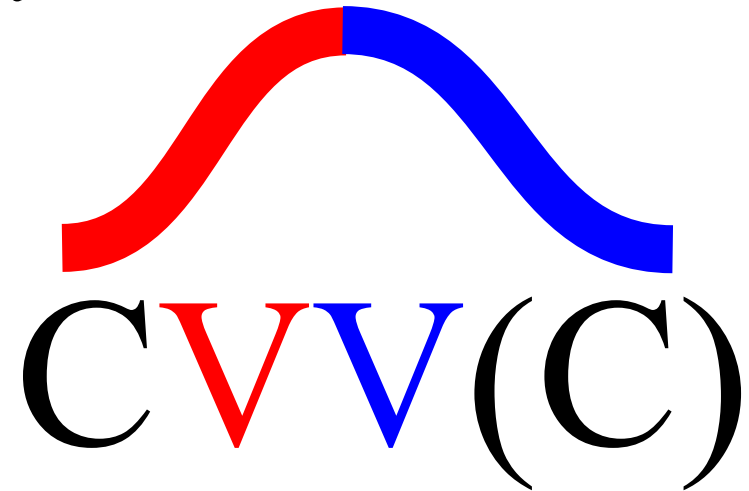
Brady Dailey - Boston University

5. How does a pitch-accent turn into a level tone?

The Proto-Pomo accent is perceived w/a **low-falling pitch** on **CVV(C)** and **high-level pitch** on **CV(C)**. This is reanalyzed as **H** on accented **CV(C)** vs. **L** on accented **CVV(C)**. Tonal minimal pairs are created through subsequent sound changes that alter the historical syllable structure.

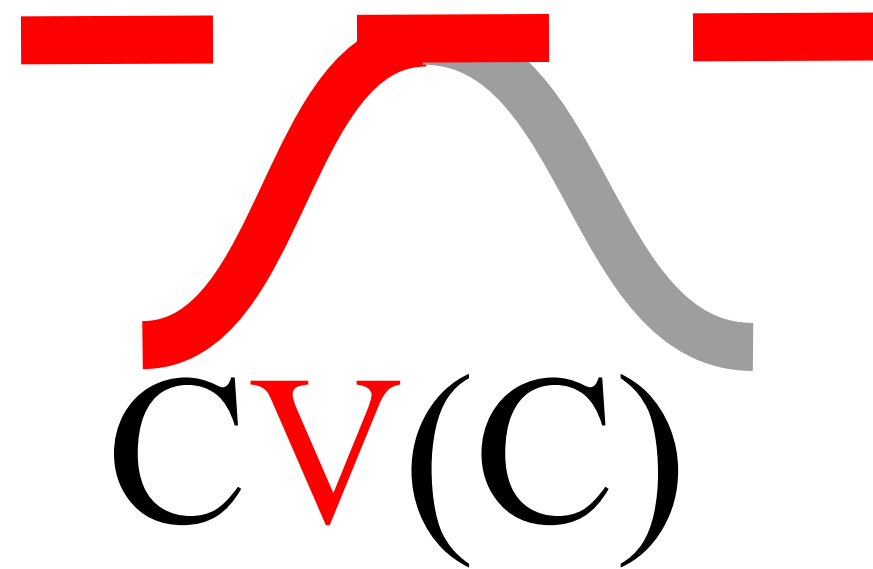
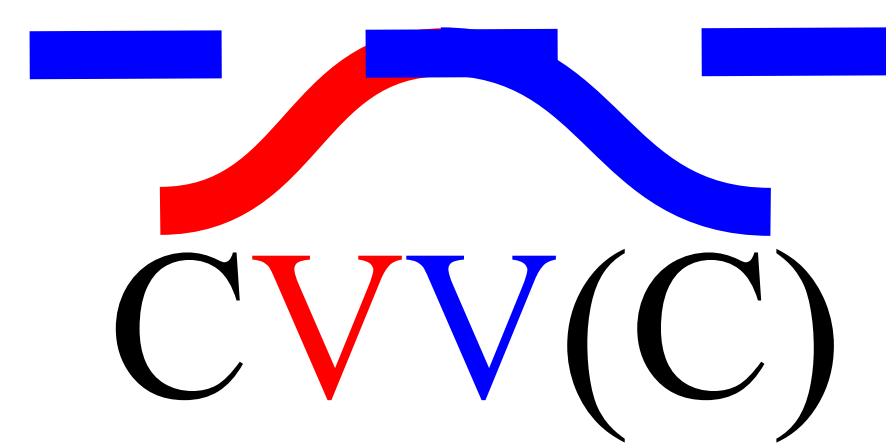
a. Segmental anchoring: Pitch movements are anchored to specific landmarks in the segmental string (Ladd 2006, i.a.). A **rising-falling** accent could be anchored so the pitch peak is consistently realized in the accented vowel and the low is reached after the accented syllable.

In accented CVV(C) syllables the fall would pass through the end of the long accented vowel. In contrast, in accented CV(C) syllables the fall would pass outside of the accented syllable or through its coda.

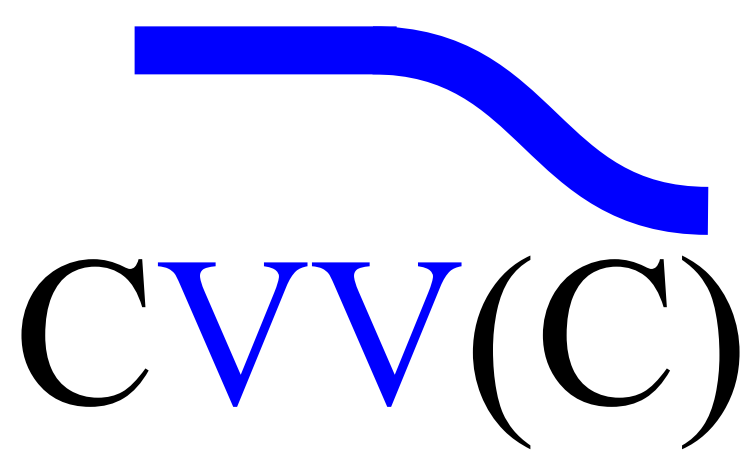


b. Perception of pitch height: Perceived pitch height is more heavily affected by f0 values at the end of a pitch movement than those at the beginning (Rossi 1978; Barnes et. al. 2024) AND it is more heavily affected by f0 values in vowels than those in consonants (Barnes et al. 2024).

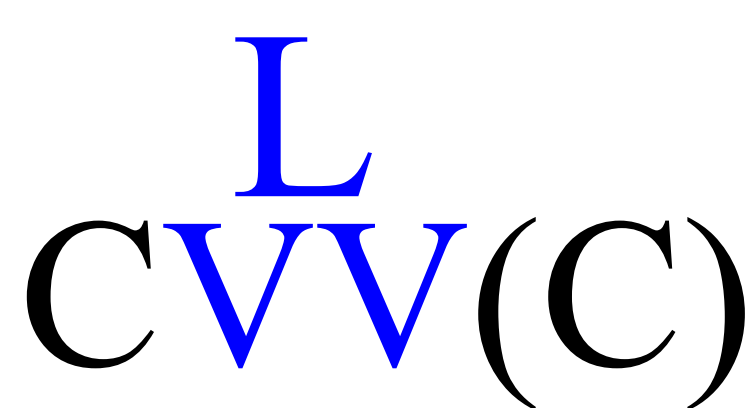
- **Low f0 values at the end of a pitch movement more heavily impact the perception of pitch height when they are realized in a vowel than when they are realized in a consonant.**
- **Therefore, the accent sounds lower in CVV(C) syllables and it sounds higher in CV(C) syllables**



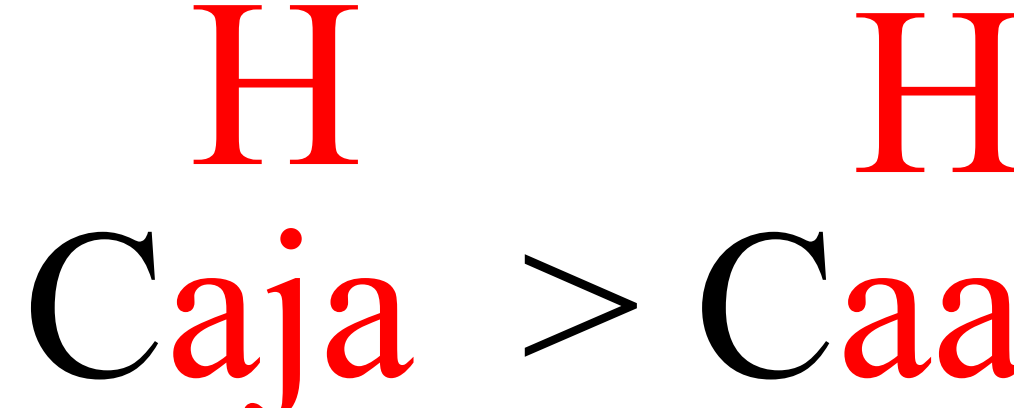
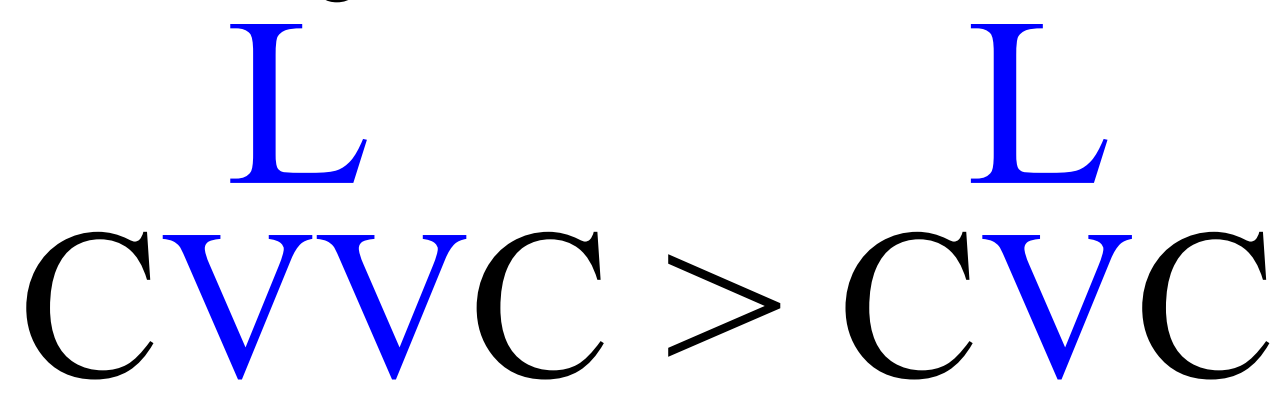
c. Perception of pitch movement: f0 movements are more likely to be perceived as **contours** on **long vowels** and more likely to be perceived as **level-pitch** on **short vowels** (House 1990).



d. Pitch-accent Reanalysis: The *perception* of a **high-level pitch** on accented **CV(C)** syllables could be analyzed as the realization of a **H** tone target, while the *perception* of a **low-falling pitch** on accented **CVV(C)** syllables could be analyzed as the realization of a **L** tone target.



e. Development of tonal contrasts: At this point, tone is conditioned by vowel length. This is why so many contrasts have a **H** on **CV** vs. a **L** on **CVV**. However, minimal pairs are formed as the contraction of **-aja* creates long vowels that did not develop **L** and the shortening of **CVVC** to **CVC** creates new short vowels retaining their historical **L**.



Summary

- **Northern Pomo, a dormant indigenous language of Northern California has developed level tone contrasts through the reanalysis of a pitch-accent.**
- **Unlike the commonly attested and well understood tonogenetic process of segmental contrast neutralization, there is no clear model through which pitch-accent reanalysis could develop into a contrast between H and L.**
- **This reanalysis is driven by the differential effects of vowels and consonants on pitch perception which could lead a rising-falling accent to sound high on a short vowel and low on long vowels.**
- **After the association of a H tone to CV(C) and a L tone to CVV(C) is established, subsequent sound changes create new long vowels and shorten old long vowels, creating minimal pairs and rendering the relationship between phonemic vowel length and tone unpredictable.**

6. What is the evidence that pitch-accent led to tonogenesis in Northern Pomo?

There are 7 Pomoan languages; none have been described with lexical tone.

Northern Pomo is most closely related to Southern, Kashaya and Central Pomo, forming a Western Pomo Branch (Oswalt 1964).

346 item comparative word list in McLendon’s (1973) *Proto-Pomo* updated to reflect Northern Pomo’s tone based on Vihman and O’Connor’s fieldwork.

- **94% N. Pomo L** tone corresponds to **CVV(C)** syllables in Proto W. Pomo.
- N. Pomo **H** bearing **CVV** syllables correspond to *-aja* in Proto W. Pomo.
- N. Pomo **H** bearing **CVC** syllables correspond to **CVC** in Proto W. Pomo.

Gloss	Northern	Southern	Kashaya	Central	Proto W. Pomo	Proto Pomo
THING EARTH	mà:	ʔam:a	ʔama:	'ma:	ʔa'ma:	'ʔa(:)ma: ʔa'mat
SOUR	mòts'	mo:ts'	mo:s'	'mo:s'	'mo:ts'	'mo:ts'
YOU PL NOM	má:	ʔa:maja	ʔamaja	'maja	ʔa'maja	ʔa'maja
ROTTEN	tʰót'	ʔihtʰot'	htʰot'	'tʰot'	ʔih'tʰot'	--

Proto-Pomo accent: McLendon suggests that the Proto-Pomo accent was a “pitch-stress accent”.

- Evidence from Eastern Pomo (McLendon 1975) and Kashaya (Oswalt 1961) suggests that it may have been realized with a **rising-falling** contour.
- **92% of the *CVV syllables are reconstructed to accented positions in the Proto-Pomo word-list.**

The Proto-Pomo accent is involved in the development of N. Pomo L tone

Northern Pomo’s tonogenesis proceeded in 3 steps:

- (1) Pitch-accent is reanalyzed as **H** on **'V** and **L** on **'VV**
- (2) **-aja* sequences contract into **a:** creating tonal minimal pairs on VV
- (3) Some **CVVC* shorten to **CVC** creating tonal minimal pairs on CVC

Proto-Pomo accent as lexical stress + phrasal pitch-accent:

- Developments in prosodic phonology (Gordon 2014, i.a.) suggest that the Accent’s pitch contour may be a phrasal pitch-accent and not just a phonetic correlate of lexical stress.
- Kashaya’s accent is associated with the “tone phrase” not the word.

N. Pomo’s tonogenesis may involve the reanalysis of a phrasal pitch-accent into the lexical representation of some words!

7. Further Questions?

- Pitch-accent has been implicated in the development of word-level pitch contrasts in languages like Swedish, Serbo-Croatian and Franconian. Investigating the phonetic and perceptual mechanisms involved in the development of these languages could further our understanding of word-level prosodic typology.

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